ELSEVIER

Fire Safety Journal **24** (1995) 131–148
© 1995 Elsevier Science Limited
Printed in Northern Ireland. All rights reserved
0379–7112/95/$09·50

0379-7112(95)00019-4

A Computer Model for the Evacuation of Large Building Populations

Peter A. Thompson & Eric W. Marchant

Fire Safety Engineering Group, Department of Civil and Environmental Engineering, The Crew Building, The King's Buildings, University of Edinburgh, Mayfield Road, Edinburgh EH9 3JN, UK

(Received 20 February 1995; revised version received 20 April 1995; accepted 26 April 1995)

ABSTRACT

The computer model 'SIMULEX' is designed to simulate the escape movement of thousands of individual people through large, geometrically complex building spaces. The model is intended for use both as a research and design tool to analyse the evacuation of large populations through a wide range of building environments.

The computer program assigns a variety of attributes to each individual in the building population. These attributes include a co-ordinate position, angle of orientation, and a walking speed for each person. Specific algorithms that facilitate the simulation of escape movement include distance mapping, wayfinding, overtaking, route deviation, and adjustments to individual speeds due to the proximity of crowd members. These algorithms contribute to a computer package that displays the building plan and the position and progress of individual building occupants as they walk towards, and through the exits.

1 INTRODUCTION

The demands of modern building design have brought about the development of building spaces with increasingly larger and more geometrically complex areas. These types of building include department stores, concert venues, and shopping malls where one of the optimal design objectives is to maximise the size and shape of a

building plan with respect to the size of population that can be accommodated. In such buildings, the overall geometric form and spatial complexities may exceed the normal design constraints specified by the statutory regulations[1].

In certain buildings, a suitable modelling or simulation method may be required in order to analyse how the complexities of the building layout may effect a potential evacuation of the occupants. Most of the available computer programs for modelling crowd motion and the evacuation process are of the 'network–node' type, which make large assumptions about the movement of people through building spaces, as '*en masse*' unified crowd flows.

Models such as EXIT 89[2] use mathematical formulae for flow rates and average speeds of groups of people, based on real-life observations of crowd motion. EXIT 89, like most network–node models, assigns all people in a room 'node' the same walking speed, based upon the occupant density in that room. Some recent packages, such as EXODUS,[3] EGRESS,[4] and that described by Toshiyuki[5] attempt to simulate the movement of individual persons through aircraft, building structures or transportation vehicles, by using a fine network of nodes which segment a single, defined, area into many linked nodes. These have not led to any significantly new developments or a better understanding of the mechanisms by which individuals interact with each other to produce the process known as 'crowd movement'.

The development of the computer program SIMULEX is the first attempt to model the escape movement of each individual occupant through a building, where the walking speed of each person is assessed independently of the average density of a group in a defined area of a building. This approach allows each individual to slow down if another, slower, person presents a potential obstruction to forward movement. Other actions, such as overtaking and 'body-twisting,' are then assessed. The overall effect is to simulate crowd movement, by allowing each person in the crowd to decide upon their own walking speed and direction of travel, depending upon the proximity of other individuals and the shape of the surrounding building structure. This allows the user of the model to look more closely at the causes of crowd congestion and the complex interactions that might occur when groups of escaping people 'merge' as they meet in the corridors or pathway junctions of a building structure.

This paper describes Version 2.0 of the SIMULEX package which only models the movement of people on a single, level surface. However, the development of Version 3.0 for multi-level buildings and staircases is now underway.

2 THE DEFINITION AND SPATIAL ANALYSIS OF A BUILDING PLAN

Two purpose-built programs 'DRAWPLAN' and 'GRIDFORM' handle the input and spatial analysis of a building plan, creating data files for subsequent use by SIMULEX to facilitate the accurate definition of a building space. All of the programs are written in Microsoft C7/C++.

2.1 To input the building plan with 'DRAWPLAN'

DRAWPLAN is a simple drawing package that allows the user to input a two-dimensional building plan. The program is intended only to be used to create files that will be used by SIMULEX. It is intended for quick and simple input of a building plan and will only be discussed briefly because in later, three-dimensional versions of the SIMULEX package, the building plan may be defined by output from a commercial CAD package.

DRAWPLAN allows the user to create a building plan by defining the building shape as a series of wall units, as illustrated in Fig. 1. Each

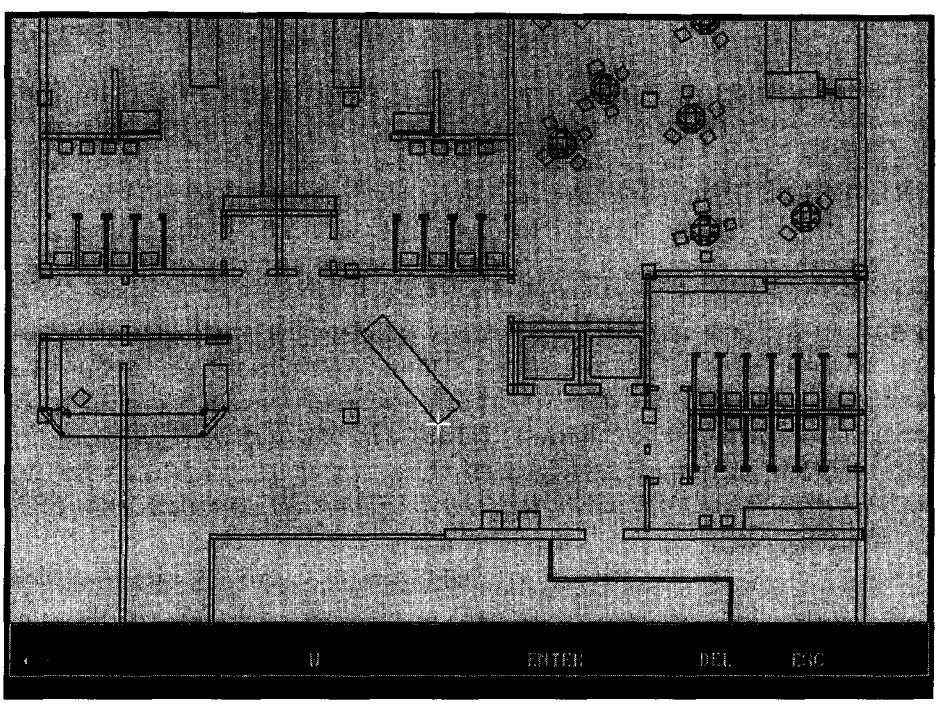


Fig. 1. A sample screen display from DRAWPLAN.

wall unit is a rectangle, whose length and orientation are defined by specifying two points with the computer 'mouse'. The width of each wall unit is also specified by the user. There is no specific facility to draw curves. Curved shapes can be closely approximated by drawing a series of short wall units that are linked and are rotated at gradually changing angles.

This method of input allows a complex building plan to be specified in a fairly short space of time. When a building plan is complete, the file containing the corner co-ordinates of each wall unit is saved onto disk for subsequent use by both GRIDFORM and SIMULEX.

2.2 'Distance mapping'—a method of spatial analysis

The computer program 'GRIDFORM' takes the building plan data file produced by 'DRAWPLAN' and uses it to generate a 'distance map' for that building. A 'distance map' is defined by a fine mesh of spatial blocks which covers the plan area of a building. The numerical value assigned to each block in the mesh is equal to the optimal 'travel distance' from the centre of that block to the nearest available final exit. This value takes account of all changes in the direction of travel required to negotiate walls and physical obstructions within the building space, along the optimal route.

The maximum and minimum x and y co-ordinates of the building plan are assessed, and are used to ascertain the dimensions of a rectangular mesh that is capable of containing the entire building plan. GRIDFORM then generates a mesh of thousands of $0{\cdot}25 \times 0{\cdot}25$ m square spatial blocks in order to represent the total building plan area. The numerical value assigned to each block is an integer that defines whether the spatial block that it represents is part of a solid object (value $= -2$), or represents open space (value $= -1$). All spatial blocks that contain exits which represent a 'final destination' are assigned a numerical value of zero, denoting that these areas have a travel distance to exit of zero. The default setting in the current version of GRIDFORM automatically assigns zero values to all blocks at the perimeter of the spatial mesh, representing lines of 'complete exit' from the building plan.

After the 'zero' exit values have been assigned, GRIDFORM applies a sequence of 'multiplied-distance' algorithms to the mesh. The functions involved in the application of these algorithms are mathematically complex, but the basic principles are as follows.

(a) All non-negative numbers in the mesh are identified as 'base'

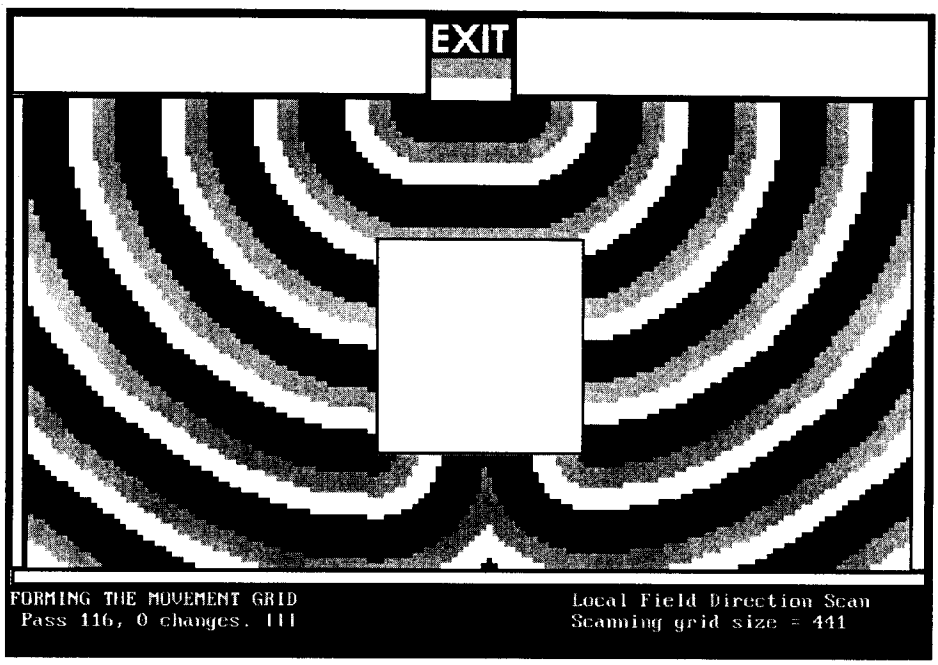


Fig. 2. Screen display of a simple distance map for a 32·0 m × 22·5 m area. Note: each shaded band represents a 0·8 m 'contour'.

blocks; that is, all spatial blocks that have been assigned a positive travel distance.

(b) 'Base' blocks that are in close proximity to 'open space' blocks which possess the value −1, are identified for calculation.

(c) The value of optimal travel distance for each 'open space' block in close proximity to a 'base' block is calculated. The travel distance is calculated by adding the 'travel distance to exit' contained in the mesh by the 'base' block, to the distance from the 'base' block to the 'open space' block. In short, travel distances are calculated by stepping from one block to another, and summing the increase in distance incurred by the 'step' each time.

(d) Steps (a)–(c) are repeated thousands of times until no further changes are made to the mesh values, and all optimal travel distances have been calculated.

When the distance map is complete, it is compiled as a data file and saved onto disk for further use by SIMULEX.

Samples of distance maps are illustrated in Figs 2 and 3. The values of travel distances are represented in bands of numerical values, and shaded as 'contours' of distance from the exits. The application of the 'distance mapping' technique allows the user to assess a total building space, in terms of travel distance and final exit where the only input required is a file containing the positions of walls and obstructions within the building envelope.

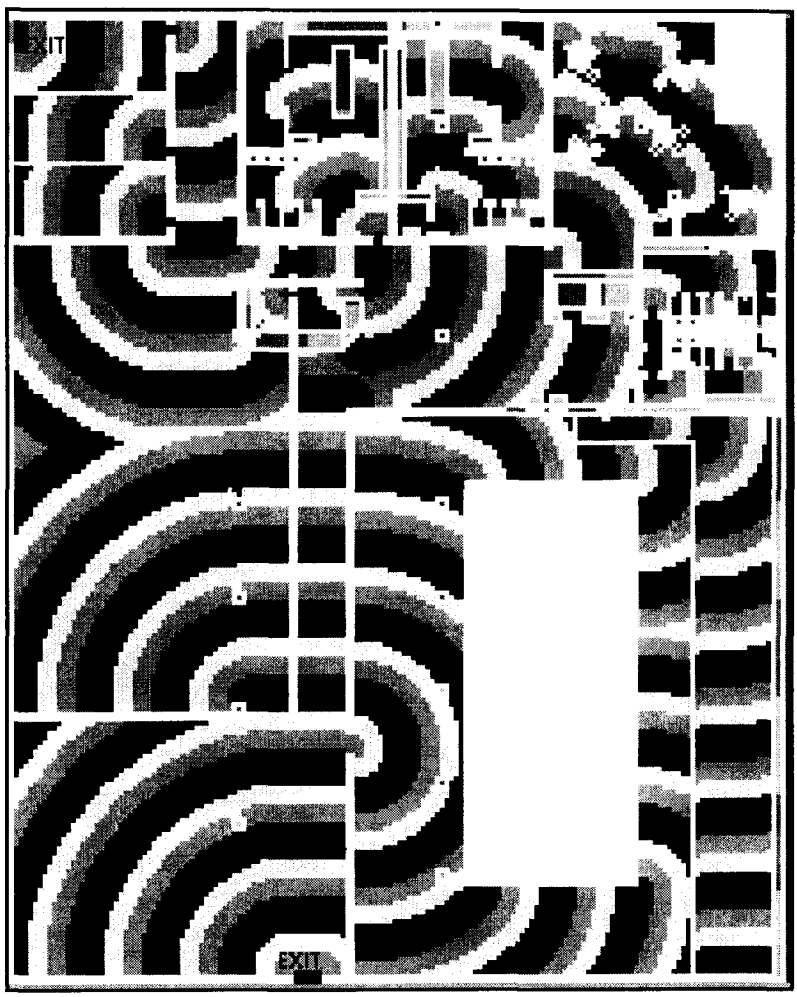


Fig. 3. Screen display of a distance map, for a 40·8 m × 52·2 m area. Note; each shaded band represents a 1·0 m 'contour'.

3 INITIAL STAGES IN THE EXECUTION OF SIMULEX

When SIMULEX is executed, the first input that is required from the user is the project name which has been assigned to the files created by DRAWPLAN and GRIDFORM. These files are loaded from disk, and locked into memory for access by all of the function routines in SIMULEX.

3.1 Assessing the optimal route to exit

The first information screen presented by SIMULEX uses the pre-defined building plan and the distance map to display information about travel distances. The two options available to the user are 'Point Travel Distance' and 'Maximum Travel Distance'. The first option, 'Point Travel Distance', allows the user to specify any position on the building plan with the computer mouse. The second option 'Maximum Travel Distance' scans the entire distance map to find the highest value, which therefore represents the position in the building that is most remote from any exit. When the initial starting position has been defined by either option, SIMULEX displays the total travel distance from that point (extracted from the distance map) and then uses wayfinding techniques to plot the optimal escape route out of the building.

The program plots the route by generating a line from the 'start' point, and progressing in linear steps of 0·25 m. The direction of travel for each step along the route is reassessed at each step point. At each step position, SIMULEX scans the surrounding 21×21 spatial blocks on the distance map, and calculates an immediate angle of travel which, if drawn on a distance map, points perpendicularly across the contours illustrated in Figs 2 and 3. The angle of travel is accurate to $\pm 2{\cdot}6°$. This accuracy is maintained when the angle of travel is reassessed after each step, and allows the plotted route to negotiate obstructions with a high degree of geometric complexity.

The x and y positional co-ordinates (in metres) for an object or person in the building are defined by floating point numbers with 7 significant figures, which allows geometrically accurate route calculations. The 0·25 m resolution of the distance map does not dictate the accuracy of the route calculations, because the only two functions that the distance map performs are: (a) to yield a value for total distance to exit, and (b) to enable the immediate optimal angle of travel to be assessed from any point in the building.

The travel distance from the initial starting point is displayed in metres, and is obtained from the value contained by the distance map at

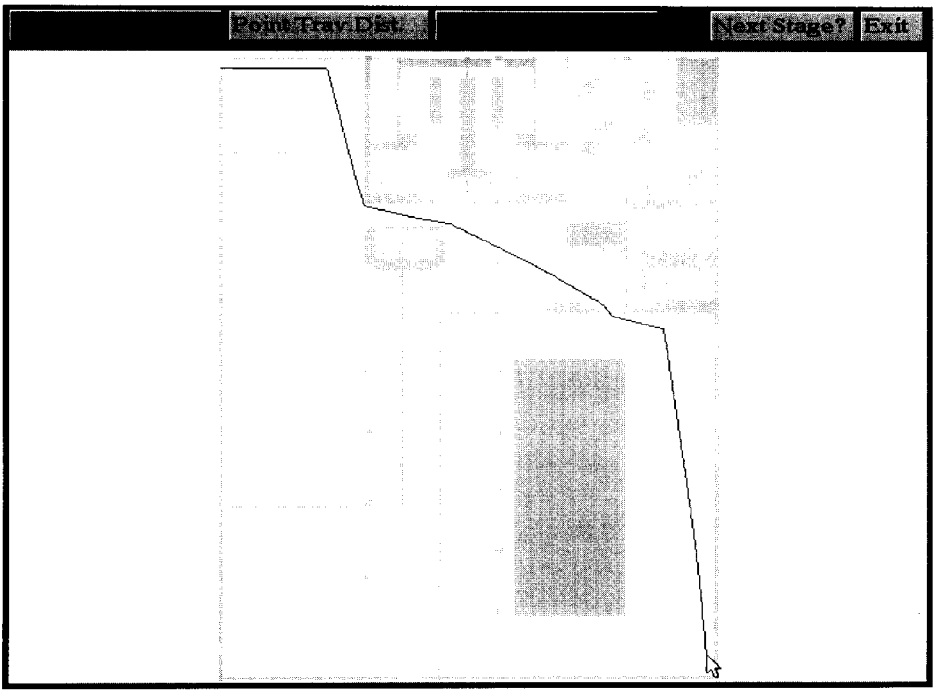


Fig. 4. Screen display showing the assessment of travel distances.

that point. The typical inaccuracy of travel distances is 0 to +3% of the total length, in addition to the possible inaccuracy of 0–0·25 m when locating an accurate floating point position within the 0·25 m × 0·25 m spatial blocks at the origin and final destination points of the route.

3.2 The definition of the building population

SIMULEX allows the user to define the occupancy in different areas of the building plan. The user can change the scale at which the building plan is viewed on the screen, and can move around to different areas and rooms. At any location, the user defines a populated zone by moving the mouse and clicking onto the points of an area (square, rectangular, or any polygonal shape). When the co-ordinate points have been specified, the total area of the defined polygonal shape is calculated.

When a populated zone has been defined, and the area calculated, the user is asked for the density of the population that is to occupy that section of the building space. A grid of occupants, with regular spacing between each person is then located within the available space of the

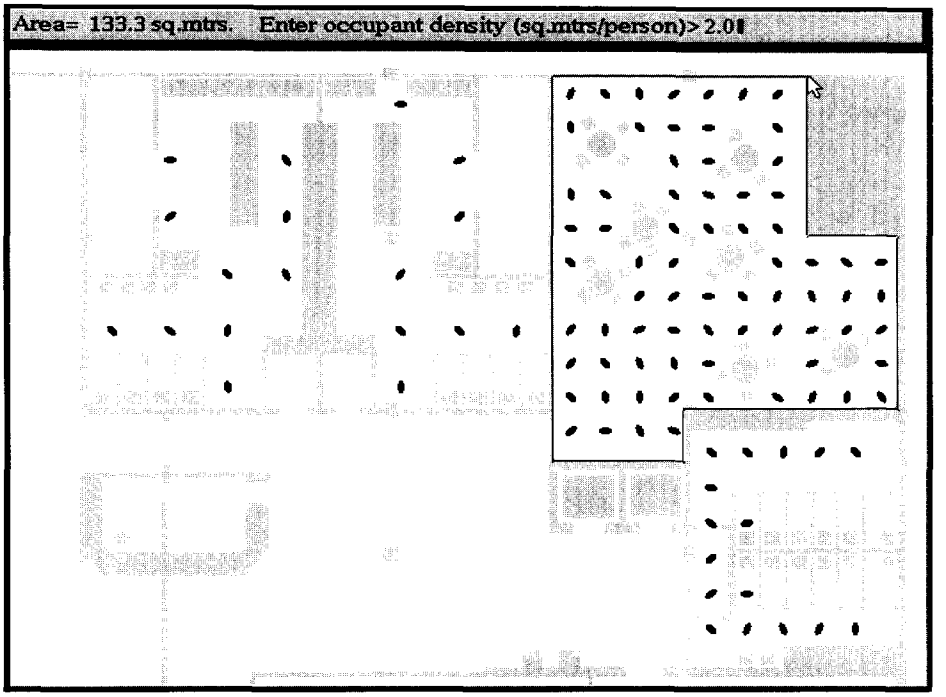


Fig. 5. Screen display showing the specification of a populated area.

populated area, as shown in Fig. 5. When a population grid has been calculated, the occupants' co-ordinate positions are stored in memory. Each person is randomly assigned an initial angle of orientation and a normal, unimpeded walking speed of between 0·8 and 1·7 m/s. This is intended to represent a population that contains an even distribution of males and females, with ages ranging from 12 to 55 years, based on data from Ando *et al.*[6]

When the user specifies that the definition of the population is complete, the program processes the population figures, and arranges the data into 'stacks', in order of how far each person is away from the nearest exit. These data stacks are then saved as a population file onto disk.

3.3 Body shapes and inter-person distances

The normal approach to defining body sizes and shapes in evacuation models is to use the form of 'body ellipse' defined by Fruin[7] and Predtechenskii and Millinskii.[8] This remains a very useful way of representing the shape of the human body, from a plan view. However, the use of an elliptical form gives rise to very complex calculations

when assessing whether or not two bodies, of known orientation and in close proximity to each other, are in contact. The point(s) of contact (or intersection) can be assessed by evaluating the fomulae for two rotated ellipses, which are a known distance apart, but the calculation process requires that the equations be solved by a method of iteration, such as the Newton–Raphson method.[9] This becomes unnecessarily complicated, considering that the elliptical form is only an approximation to the physical shape of a body. Therefore, SIMULEX uses a different mathematical form to represent the shape of a human body, by using three circles (one for the main body and two for the shoulders). This form is illustrated in Fig. 6.

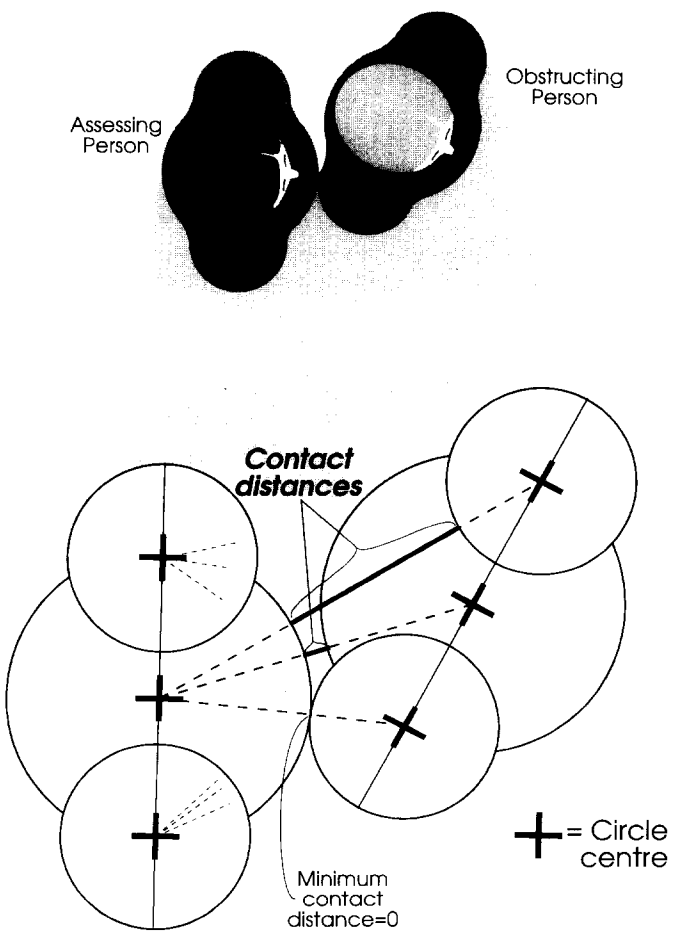


Fig. 6. Mathematical representation of two human bodies, and the assessment of contact distances.

The 'three circle body' uses very simple calculation principles to assess bodily contact. The assignment begins by calculating the nine 'contact distances' between the two sets of three circles, representing the bodies. For two circles, the 'contact distance' is defined as the total distance between the circle centroids minus the sum of the two circle radii. The most valuable 'contact distance' is the 'minimum contact distance', which represents the smallest distance between the physical envelopes of the two bodies. When the 'minimum contact distance' is equal to zero, two bodies are in contact. However, even when the bodies are not in contact the calculation is still valuable to SIMULEX because the 'minimum contact distance' is used as the 'inter-person' distance, which is used at a later stage in the calculations to predict the impeded walking velocity of an 'assessing person'.

In version 2.0 of SIMULEX each person is assigned the same average body size, but it is intended that each person will be assigned different body dimensions, depending on age and gender. The average body size used by this program is $0{\cdot}5\,\text{m} \times 0{\cdot}3\,\text{m}$, which is an average of the data presented by Pauls,[10] Fruin[7] and Predtechenskii and Millinskii.[9]

4 MODELLING EVACUATION: TIME, DECISIONS, AND MOVEMENTS

The evacuation of a large number of individuals from a building can be modelled when: the building plan has been entered; the 'distance map' has been generated; and the population defined. SIMULEX is capable of modelling buildings with plan sizes in excess of $50{,}000\,\text{m}^2$, containing a maximum population of 15,000. However, if more than 2000–3000 people are modelled, the computer processing time required for total evacuation will be in excess of 5–6 hours on a 486 DX2-66 PC.

The whole process of evacuation occurs in time-steps of $0{\cdot}1\,\text{s}$. The figure of $0{\cdot}1\,\text{s}$ was used because over this period of time, the fastest person will only move forward $0{\cdot}17\,\text{m}$, which is only slightly more than half the body depth of one person. This prevents the possibility of a person 'stepping through' a solid obstruction without his/her body encountering the presence of that obstruction. It produces an acceptable time for overall computer processing, while maintaining the accuracy that is necessary for the simulation of individual movement. The computer processing time required for the complete decision making procedure, movement data assessment system and graphic display for each individual, at each time-step is approximately $0{\cdot}009\,\text{s}$, when executed on a 486 DX2-66 PC.

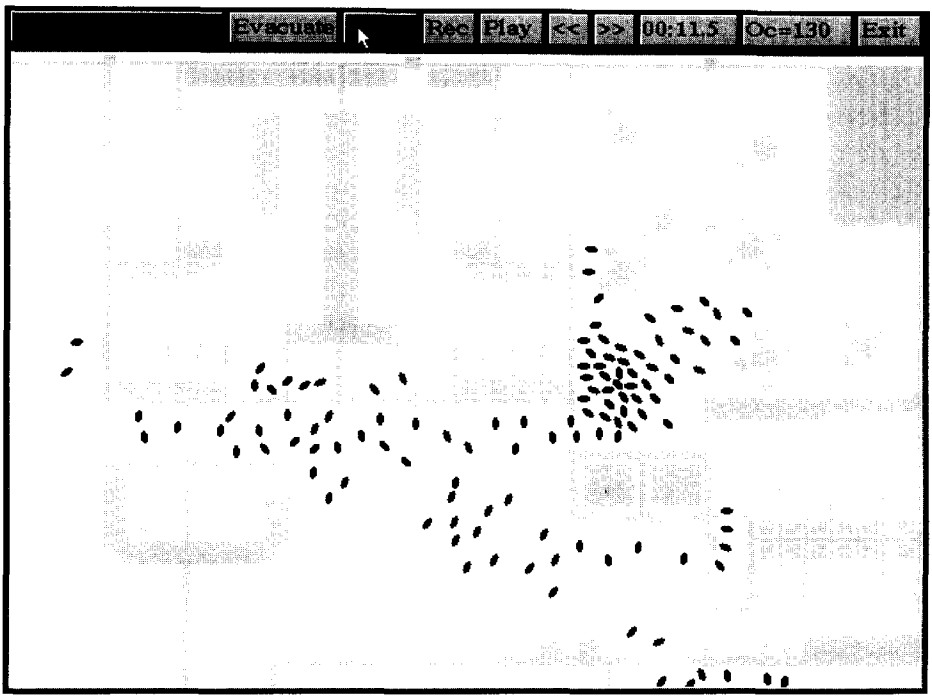


Fig. 7. Screen display from SIMULEX for a sample evacuation.

A screen display from SIMULEX, showing the visual display for a simulated evacuation, is presented in Fig. 7. The whole evacuation simulation consists of a series of repeated analytical loops. At each time-step, the position and attributes of each individual are retrieved, and a set of decision/movement processes are executed which predict the immediate parameters of motion for that person. The new position, angle of orientation, and distance to exit are then calculated and stored in computer memory. The processing for the whole population occurs sequentially in the order of the person nearest to exit first, to the person furthest from exit, last. The order of data storage reflects this dependency on 'distance to exit'.

When all individuals have been assessed for one time-step, the order of data storage is reassessed to ensure that all data stacks conform to the specified order in terms of the distance to exit of each occupant. When the data stacks have been reassessed, all occupants possessing a distance to exit of zero are removed from the stacks, erased from the screen display, and are deemed to have left the building. The 'Number of Occupants' display reflects this change in the number of building occupants. The next time-step is then processed in the same way.

The analytical loop of:

execute decision and movement process for all occupants
reassess order of data storage
remove all occupants with a distance-to-exit of zero
update the time and number of occupants
– – – – – repeat the whole process – – – – –

is repeated until the number of occupants in the building is exactly equal to zero. Thus, the entire evacuation of the building is modelled by SIMULEX by combining algorithms which carry out a number of very different processes. Most of these algorithms concentrate on the physical aspects of escape movement, but further developments will incorporate algorithms to emulate some of the complexities of human psychological response, and the mutual interaction that occurs between individual occupants.

The procedure carried out by the primary algorithms for the movement of each individual at each time-step are described in Sections 4.1–4.4.

4.1 The assessment of initial 'optimal' direction of travel

The floating point co-ordinate position of each person is retrieved from the data stacks. This position is then translated to an integer form, based on a multiple of 0·25 m, and located on the distance map. The values of the surrounding 441 spatial blocks are analysed, in the way described in Section 3.1, and an angle is produced which points in a direction that is at right angles to the surrounding distance contours. This angle (θ_0) represents the ideal angle of forward movement for a person who seeks the most direct route to the nearest exit. It is regarded as the initial 'optimal' direction of travel, and is the first parameter of motion to be calculated for each person, at each time-step.

The calculation of θ_0 is carried out by the only algorithm that uses an integer form of the co-ordinate position of a person. All subsequent algorithms use the full floating point form, where the exact position of the centre of a person's body is expressed with an accuracy of greater than 1 mm.

4.2 Assessing individual walking speeds

In previous analyses of evacuation movement, the calculations for speeds of movement relate to the motion of groups of people, and the average crowd densities. The figures quoted by Fruin,[7] Predtechenskii

and Millinskii,[8] Ando *et al.*,[6] Polus *et al.*,[11] and Hankin and Wright,[12] form the basis for such calculations. However, the specific movement of people walking within a crowded area of a building has not previously been analysed in sufficient depth to ascertain the parameters of movement for each individual person. Therefore, a number of tests were carried out at different sites in Edinburgh, where the egress movements of groups of people from buildings were filmed with a video camera. The filmed footage was processed using a series of image analysis techniques to assess the primary parameters for the movement of individuals within crowded areas. Further tests and analyses will be carried out, and the method of analysis and results will be the subject of a further publication. Preliminary results are contained in the Honours thesis by Ricketts.[13]

The analysis of the footage, filmed at Edinburgh, has led to the derivation of a relationship between the walking speed of an 'assessing' individual and the 'inter-person distance' from the 'assessing' person to the nearest 'obstructing' person. The 'inter-person distance' is a measure of the distance between the centre points of the bodies of two individual persons. The relationship between the walking velocity of an 'assessing' person and the inter-person distance to an 'obstructing' person is presented in Fig. 8. The median values are based on a 'best fit' curve for the obtained data, and the age-weighted values are based on a combination of the median values and the data quoted by Ango *et al.*,[6] for normal, unimpeded walking speeds. The graphical form described in Fig. 8 indicates that the walking speed of individual people decreases when the inter-person distance is less than a specific value (the

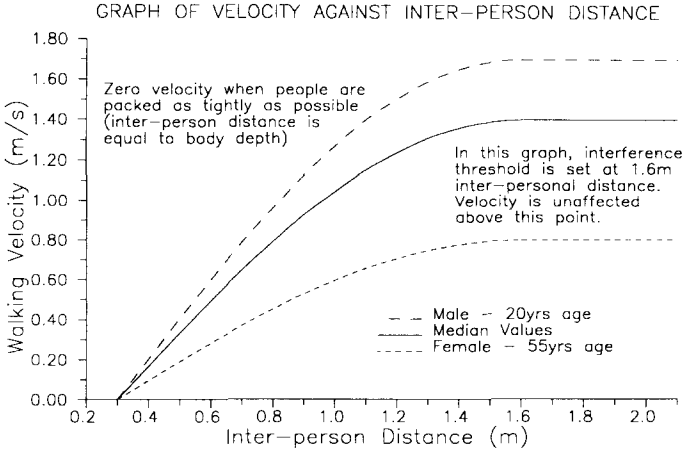


Fig. 8. The relationship between walking velocity and inter-person distance.

'threshold distance'). The walking velocity becomes equal to zero when there is no clear distance between the bodies of the 'assessing' and 'obstructing' persons. The graph illustrated relates to the movement of adults under calm, fairly 'stress-free' conditions.

SIMULEX uses the relationship presented in Fig. 8 as part of an algorithm to assess the walking velocity of each modelled person in the direction of travel described by θ_0. The relationship is slightly adapted to use the 'minimum contact distance' between individuals, rather than inter-person distance, so that the body form defined in Section 3.3 is accommodated. SIMULEX scans the forward path of an 'assessing person' to detect if any person is present, and if so, the effect of the closest 'obstructing person' is assessed. The reduced walking velocity of the 'assessing person' is then calculated.

4.2 Deviation from optimal direction of travel (overtaking)

If the walking velocity of an 'assessing person' is reduced, due to the presence of an 'obstructing person', SIMULEX assesses the possibility of overtaking. The method of assessment for two potential angles for overtaking (θ_1 and θ_2) is illustrated in Fig. 9.

The values of θ_1 and θ_2 are calculated to allow the 'assessing person' to pass the 'obstructing person' with a clearance of 50 mm, to take account of body sway. This figure for 'body sway' is based on data from Bryan,[14] describing motion in crowded conditions. Either θ_1 or θ_2 will be selected as the new direction of travel if their selection allows the 'assessing person' to walk more quickly than in the original direction,

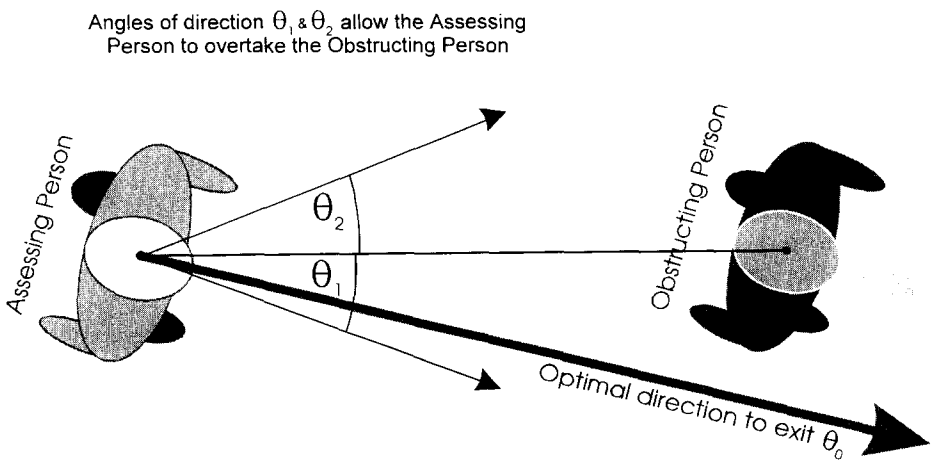


Fig. 9. The assessment of direction of travel.

by avoiding the 'obstructing person'. The preferred angle of overtake will be that which requires the least deviation from the original, optimal direction and produces a significant improvement in walking velocity. In Fig. 9, θ_1 will be the preferred direction because it requires only a small deviation from the optimal route, while θ_2 would require a much larger change in direction.

When SIMULEX is executed, it appears to produce reasonably realistic forms of overtaking when there is enough space available to the assessing person to enable this form of route deviation. The overtaking algorithm is not applied in areas where the crowd density exceeds 2 persons/m^2.

4.4 Other aspects of motion

When a modelled person comes into contact with a solid object (such as the corner of a wall unit), SIMULEX corrects the direction of travel so that the shoulder of the person is maintained at a minimum distance of 50 mm away from the face of the obstruction. The direction of travel of a person is therefore adjusted if such contact is predicted, and the resulting 'corrected' direction models the 'avoiding' behaviour of a person who would walk around a solid object. This 'clearance' is to allow for the body sway described by Bryan,[14] and is achieved by using a 'contact circle' with a diameter of 0·6 m, which acts as a spatial envelope around the centre of the modelled person. This single 'contact circle' is only used when assessing the presence of solid, immovable objects. The amount of body sway which is normally encountered when a person is adjacent to solid obstructions will be investigated further.

The rate at which the angle of orientation of a person can change, or the rate of 'body twist', is limited to 100 degrees per second, which is based on results from the image analysis tests mentioned in Section 4.2.

5 CONCLUSION

A range of new techniques for accurately modelling the escape movement of individual occupants from a building have been developed for use by SIMULEX. The 'distance mapping' technique is very useful for assessing travel distances throughout a building space. It represents the first computerised method for the accurate calculation of travel

distance and is particularly effective for the assessment of maximum travel distances in large, geometrically complex buildings.

The algorithms that control the assessment of escape routes and the deviation from such routes (in the form of overtaking) are precise, but allow the direction of travel for an individual to be reassessed when the physical presence of another person presents a potential obstruction to movement. The complexity of a total escape route has no effect on the processing time required for the calculation of the route, because only the immediate direction of travel is calculated at each time-step. The calculation of individual routes requires far less user-input than for any other contemporary program, and is also more geometrically accurate. However, each person generally walks towards the nearest exit, and the choice of exit is unaffected by signage and knowledge of the building environment. It is intended that future versions of SIMULEX will allow the user to apply weighting factors to different exits, depending on the frequency of usage and clarity of signage.

The size and shape of the body of each person is defined by using a combination of circles, which allows the potential contact of the bodies of two or more persons to be assessed accurately. However, not all of the aspects of the physical definition of each person's body have been investigated. For example, the difference in body size and shape that is synonymous with age, gender and disability needs to be explored further.

SIMULEX is the first computer model to use the relationship between walking velocity and inter-person distance, described in Fig. 8. It represents the first attempt to model the effects of the 'invasion of personal space' in relation to the movement of each individual person, and represents a form of psychological modelling. However, SIMULEX does not model other, more complex psychological aspects of escape movement such as the time taken for a person to react to an alarm; the spoken communication amongst the members of an escaping group of people; the familiarity of escape routes; and the effects of 'anxiety' and 'stress' on the occupants. It is intended that these issues be addressed in later versions of the computer model.

ACKNOWLEDGEMENT

Thanks to Geoff Mulford for his original idea of an integeric distance mapping algorithm, which was developed by the authors and forms the basis of the 'distance mapping' techniques described in Section 2.2 of this paper.

REFERENCES

1. British Standards Institution, BS 5588: Part 3. Fire precautions and the design and construction of buildings, part 3: code of Practice for office buildings. BSI, London, 1983.
2. Fahy, R. F., EXIT 89: an evacuation model for high-rise buildings. In *Fire Safety Science—Proceedings of the 3rd International Symposium,* ed. G. Cox & B. Langford. Elsevier Science, London, 1991.
3. Galea, E. R. & Perez Galparsoro, J. M., A computer based simulation model for the prediction of evacuation from mass transport vehicles. *Fire Safety J.* **22** (1994) 341–66.
4. Ketchell, N., Webber, D. M., Marriot, C. A., Stephens, P. J., Brearley, I. R. *et al.,* The EGRESS code for human movement and behaviour in emergency evacuations. In *Engineering for Crowd Safety—Proceedings of the International Conference for Crowd Safety,* ed. R. A. Smith & J. F. Dickie. Elsevier Applied Science, Amsterdam, 1993.
5. Toshiyuki, A., Prediction system of passenger flow. In *Engineering for Crowd Safety—Proceedings of the International Conference for Crowd Safety,* ed. R. A. Smith & J. F. Dickie. Elsevier Applied Science, Amsterdam, 1993.
6. Ando, K., Ota, H. & Oki, T., Forecasting the flow of people (in Japanese). *Railway Res. Rev.* **45** (1988) 8–14.
7. Fruin, J., *Pedestrian Planning and Design.* The Metropolitan Association of Urban Designers and Environmental Planners, New York, 1971.
8. Predtechenskii, V. M. & Millinskii, A. I., *Planning for Foot Traffic Flow in Buildings.* Amerind, New Delhi, 1978 (translated from original publication in Russian, 1969).
9. Minkowycz, W. J., Sparrow, E. M., Schneider, G. E. & Pletcher, R. H., *Handbook of Numerical Heat Transfer.* Wiley, Chichester, UK, 1988.
10. Pauls, J. L., *Fires and Human Behaviour,* Ch. 13 and 14, ed. D. Canter. Wiley, London, 1980, pp. 227–75.
11. Polus, A., Schofer, J. L. & Ushpiz, A., Pedestrian flow and level of service. *J. Transportation Engng,* **109** (1983) 46–7.
12. Hankin, B. D. & Wright, R. A., Passenger flows in subways. *Operational Res. Quart.,* **9** (1958) 81–8.
13. Ricketts, G., An investigation into the density and flow of people in public spaces by image analysis. Hons dissertation., Dept of Civil Engineering and Building Science, University of Edinburgh, 1994.
14. Bryan, J. L., Concepts of egress design. In *The Fire Protection Handbook,* 16th ed., NFPA, Quincy, MA, 1985, pp. 720–39.